

Derivatives Examples Gallery — Strategies, Payoffs & Key Metrics

A visual catalog for the Trader Screener project. Each example shows the core feature (the payoff shape) and the key metrics people read off it: max profit, max loss, breakeven(s), dominant Greek exposure, and the IV signal to screen for. The final two figures are the Greeks themselves — how delta/gamma/vega/theta behave, which is what professional options traders actually watch.

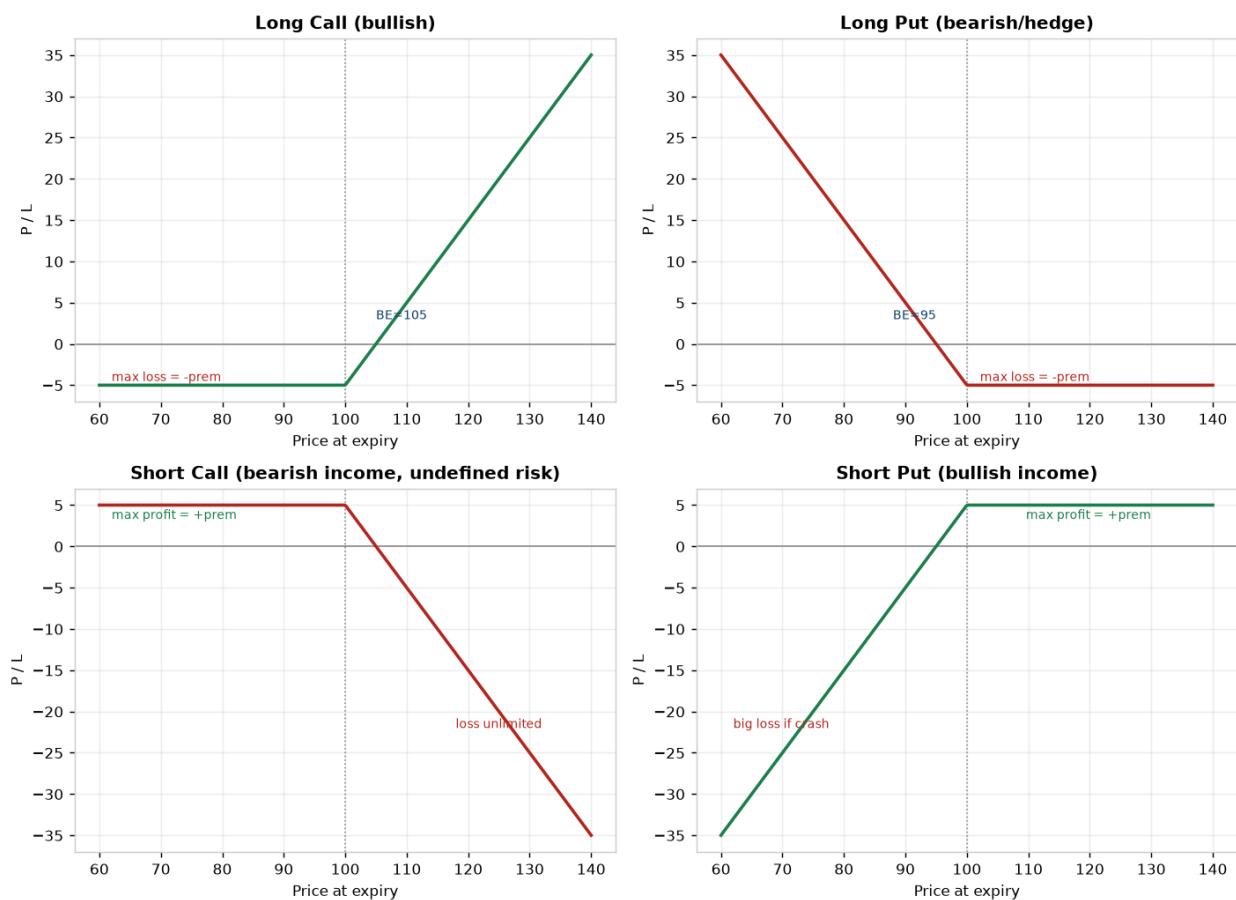
Charts generated by `_build_examples.py` (Black-Scholes for the Greeks). Illustrative params: strike $K=100$, premium $\approx \$5$, IV=25%, ~ 3 months to expiry. Shapes are exact; dollar values are examples.

How to read every options example

Five numbers describe almost any options position: 1. **Max profit** — the best outcome. 2. **Max loss** — the worst (is it *defined* or *unlimited*?). 3. **Breakeven(s)** — price(s) where $P/L = 0$. 4. **Direction (Delta)** — does it want the underlying up, down, or sideways? 5. **Volatility (Vega) + Time (Theta)** — does it *want* IV/time to rise or fall?

The recurring screen rule from the [signals doc](#): **buy premium when IV Rank is low, sell premium when IV Rank is high.**

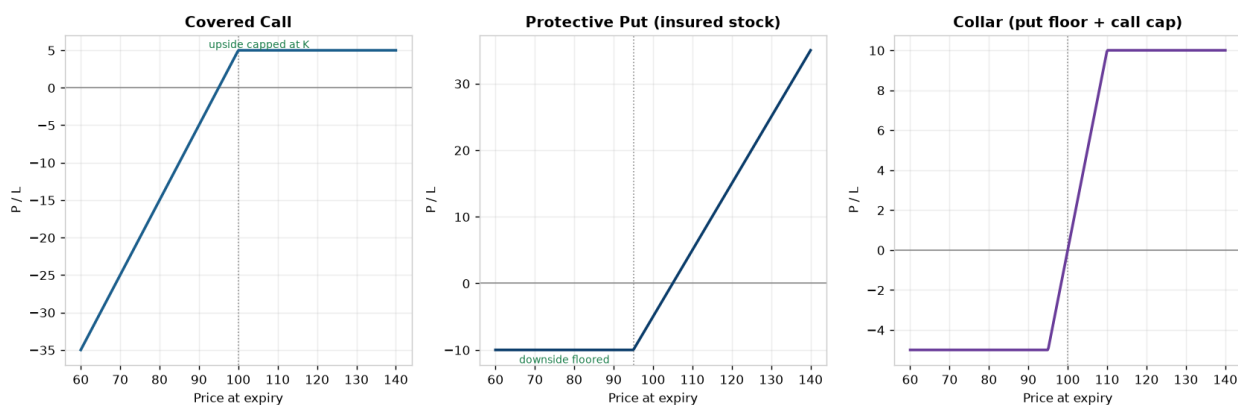
1. Single-leg positions — the building blocks



Strategy	View	Max profit	Max loss	Breakeven	Greeks bias	Screen for
Long call	bullish	unlimited	premium (defined)	K + premium	+delta, +vega, -theta	low IV Rank (cheap)
Long put	bearish / hedge	large ($\rightarrow 0$)	premium (defined)	K - premium	-delta, +vega, -theta	low IV Rank, low skew
Short call	bearish/neutral income	premium (defined)	unlimited	K + premium	-delta, -vega, +theta	high IV Rank
Short put	bullish income	premium (defined)	large if crash	K - premium	+delta, -vega, +theta	high IV Rank

Buyers are long volatility and fight time decay; **sellers** collect time decay and are short volatility. Note the asymmetry: short call risk is unlimited — sizing matters.

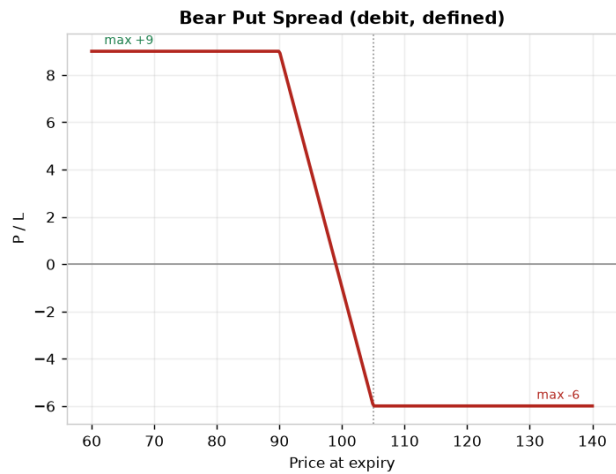
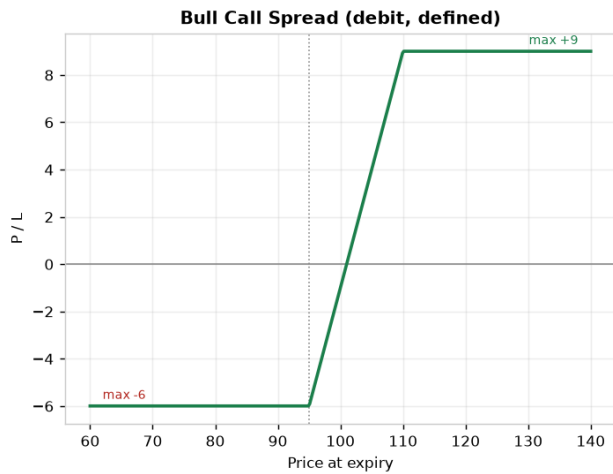
2. Stock + option — hedged & income positions



Strategy	Construction	Max profit	Max loss	Why use it	Screen for
Covered call	long stock + short call	capped at strike + premium	stock down to 0 (less premium)	income on a holding; mild bull/neutral	high IV Rank
Protective put	long stock + long put	unlimited (stock)	floored at put strike	insurance on a holding	low IV Rank (cheap insurance)
Collar	stock + long put + short call	capped (call strike)	floored (put strike)	low/zero-cost protection, range-bound	put skew rich vs call

A collar funds downside protection by selling away upside — a near-free hedge when put skew is rich.

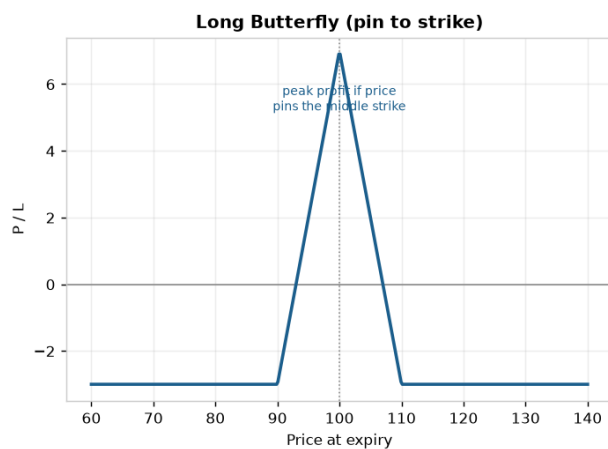
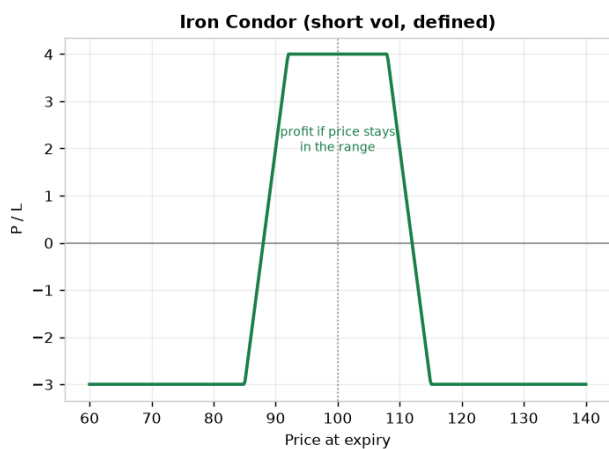
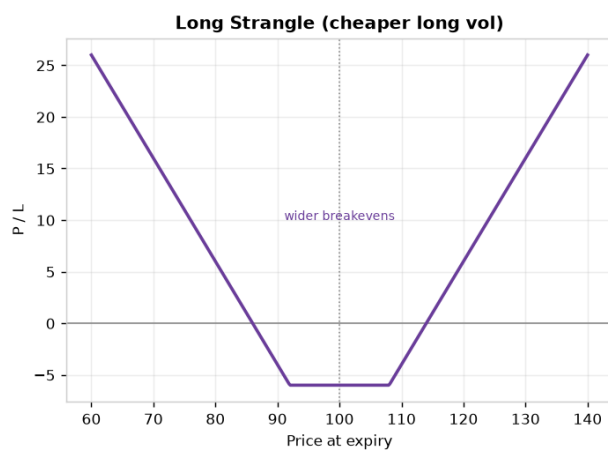
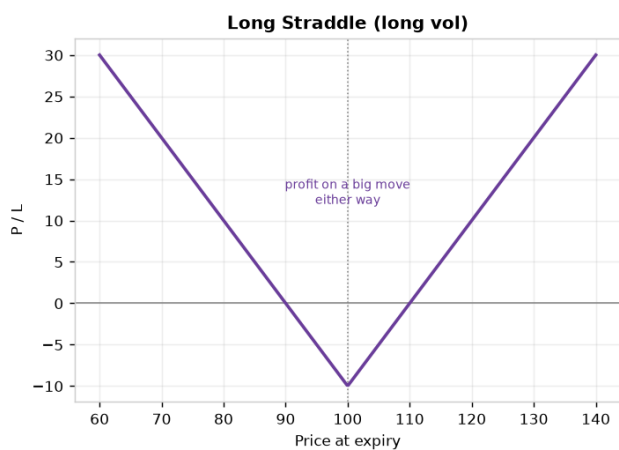
3. Vertical spreads — defined risk *and* reward



Strategy	View	Max profit	Max loss	Breakeven	Screen for
Bull call spread	moderately bullish	strike width – debit	debit paid	lower K + debit	high IV (caps vega cost vs lone call)
Bear put spread	moderately bearish	strike width – debit	debit paid	higher K – debit	high IV

Spreads trade away unlimited upside for a **lower cost and defined risk** — both legs cancel most of the vega, so they're the disciplined way to express direction when IV is elevated.

4. Volatility strategies — trading the *size* of the move, not direction

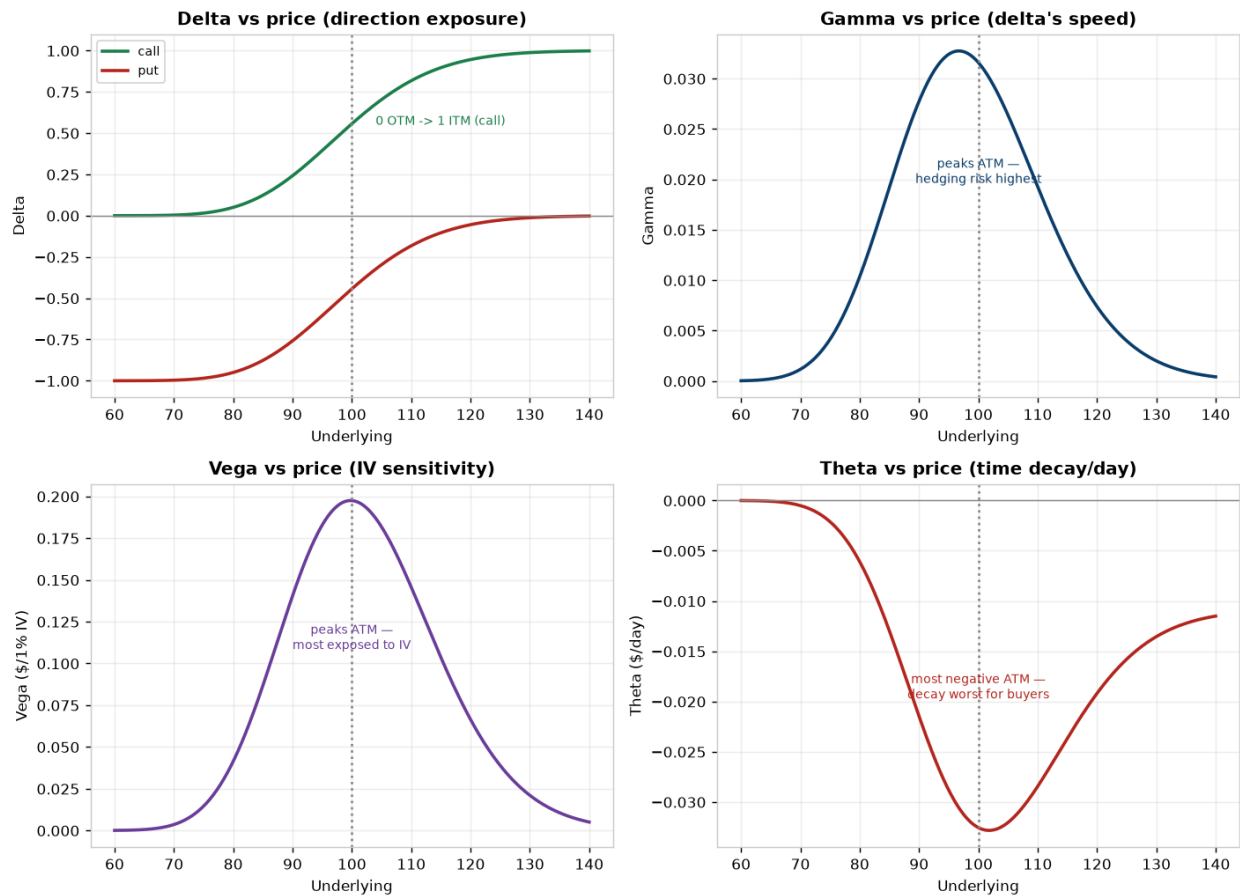


Strategy	Bet	Max profit	Max loss	Greeks bias	Screen for
Long straddle	big move either way	large (both tails)	both premiums	+vega, -theta, ~0 delta	low IV Rank before a catalyst
Long strangle	big move, cheaper	large (both tails)	both premiums (less)	+vega, -theta	low IV Rank; wider breakevens
Iron condor	price stays in a range	net credit (defined)	width – credit (defined)	-vega, +theta	high IV Rank (sell rich premium)
Long butterfly	price pins a strike	width – debit	debit (defined)	-vega near center	pin/low-movement view

Straddles/strangles are **long volatility** (want IV/realized vol to rise — classic pre-earnings buy, but beware IV-crush after). Condors are **short volatility** (want quiet markets and high IV to fade). This is the purest expression of the IV-Rank rule.

5. The Greeks — what professionals actually watch

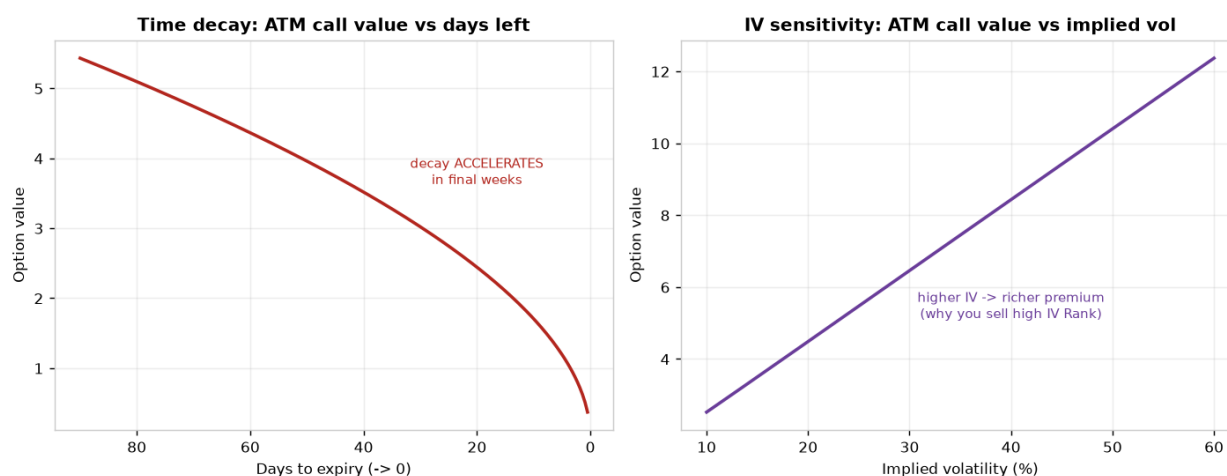
The Greeks measure how an option's value changes as price, volatility, and time move. **They are the risk dashboard** — a trader manages a book by its net Greeks, not by gut.



Greek	Measures	Key behavior (shown)	Trader use
Delta	sensitivity to underlying price	call 0→1, put -1→0 across strikes; ≈0.5 ATM	net directional exposure / hedge ratio
Gamma	rate of change of delta	peaks ATM, near expiry	hedging risk; large near-expiry ATM gamma drives dealer flows (GEX)
Vega	sensitivity to IV	peaks ATM, grows with time to expiry	volatility exposure; long options = +vega

Theta	time decay per day	most negative ATM; worst for long options	premium sellers' tailwind, buyers' drag
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Time decay & IV sensitivity over time



- **Left — time decay accelerates.** An ATM option loses value slowly far from expiry, then **decays fastest in the final weeks** (theta is non-linear). This is *why* sellers like short-dated options and buyers get hurt holding them.
- **Right — value rises with IV.** Higher implied vol → richer premium (vega at work). This is the mechanical reason to **sell when IV Rank is high and buy when it's low**.

Summary: matching action to signal

If you expect...	Action	Key Greek you're long/short	Screen signal
Up, leveraged	Long call / bull call spread	+delta	low IV Rank
Down / hedge	Long put / bear put spread / protective put	-delta	low IV Rank, low skew
Big move (any direction)	Long straddle / strangle	+vega	low IV Rank pre-catalyst
Quiet / range-bound	Iron condor / short put / covered call	-vega, +theta	high IV Rank

Pin a level	Long butterfly	–vega near center	low expected movement
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What this means for the Trader Screener

- These payoffs are **derived from the same option-chain data** as the Part-2 signals — IV, IV Rank, skew, the Greeks. Surfacing them requires the ■ **options feed** (see [SCREENER_SHORTLIST.md §2h](#)).
- A trader-grade screener should let you **filter by Greek exposure and IV Rank**, then suggest which of these structures fits (the "matching action to signal" table is the logic).
- **Unusual options activity** in any of these structures = the options-side **smart-money** signal (ties to the equity congress/insider moat).

Pair with: [Derivatives — Types & Signals](#) (the metrics in depth) and [Trading and Exchanges summary](#) (why markets price them this way).